

Experimental Verification of Automated Loss Simulation of Magnetic Components

Losses predicted by an automated magnetic simulation are checked by measuring the heat they produce

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THE PRINCIPLE

$$P_{\text{loss}} = C_{\text{Th}} \cdot dT/dt$$

Every watt lost in the core and winding of a magnetic component ends up as heat. Measuring how fast a known heat capacity C_{Th} warms up gives the real loss — independent of the simulation.

C_{Th} : total heat capacity of the stirred liquid, the enclosure and the device under test.



1 • Magnetic simulation

PLECS, Infineon SPICE models and 2D FEM in Ansys Maxwell predict the core and winding losses of inductors and transformers.



2 • Thermal measurement

The inductor runs in its converter inside a stirred, near-adiabatic liquid calorimeter; its temperature slope gives the real loss.



3 • Comparison

Buck converter, RM14LP, 400 kHz: litz within 1.5 %, solid wire measured 8–11 % above simulation.

Magnetic side: automated loss simulation

Python-based workflow: PLECS waveforms → FFT → Ansys Maxwell 2D losses of the magnetic component

- 1

Circuit model

PLECS

The converter (example: buck converter) is drawn in PLECS; the waveforms of the magnetic component are extracted.
- 2

Switch losses

Infineon SPICE

Switch losses are simulated with Infineon device models: $\approx 6.8\text{ W}$ per switch.
- 3

FFT of the current

Python

An FFT of the inductor current is done; the 10 most dominant frequency components are kept.
- 4

Magnetic FEM

Ansys Maxwell 2D

They become parallel sinusoidal sources. Winding layer stack and material (litz or solid), core size & material, insulation thicknesses and air gaps are set.
- 5

Loss result

core + winding

Total inductor loss: 6.75 W with solid wire, 6.70 W with litz wire — the quantity that is then measured as heat.

Why two tools? Each sees half of the physics

	PLECS (circuit)	Ansys Maxwell 2D (FEM)
models	whole converter, switching	fields in core and winding
inductor	ideal inductance L	real core, air gap, layers, wire
delivers	current waveform $i(t)$	core + winding losses
speed	fast	slow (mesh, small time steps)

→ PLECS supplies the current; the FEM turns it into losses.

CASE STUDY

Buck converter • RM14LP core • 400 kHz • $\approx 880\text{ W}$

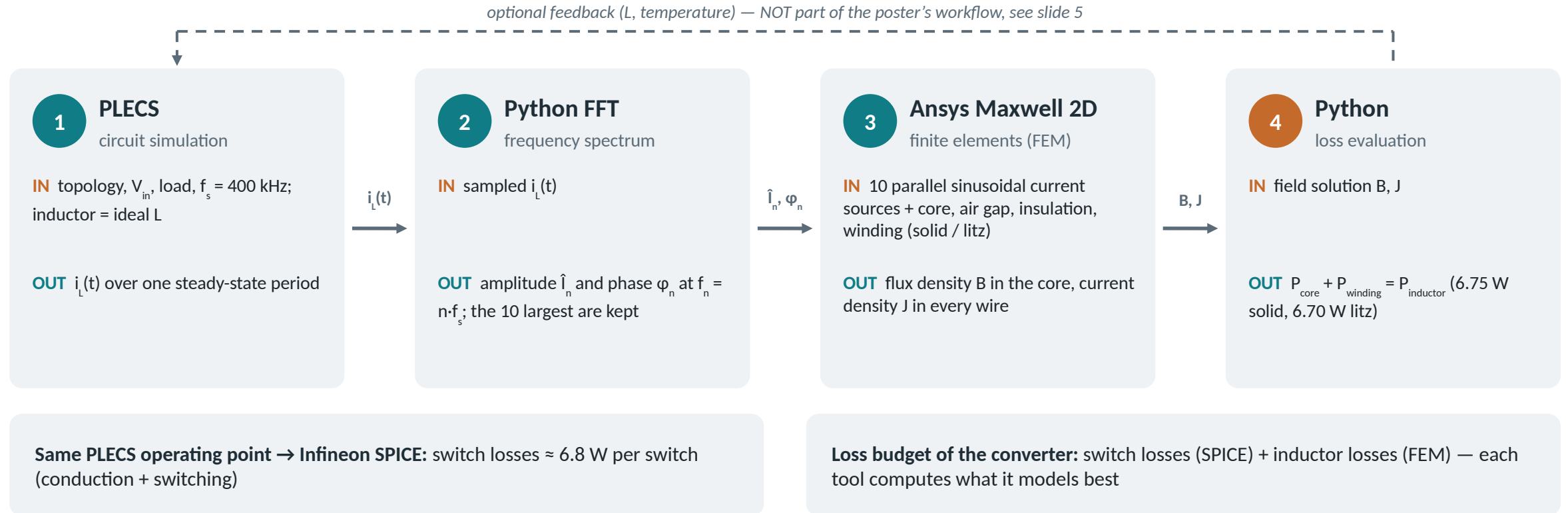
$\approx 6.8\text{ W}$
loss per switch

6.75 W
inductor, solid wire

6.70 W
inductor, litz wire

From PLECS to FEM: what is passed between the tools

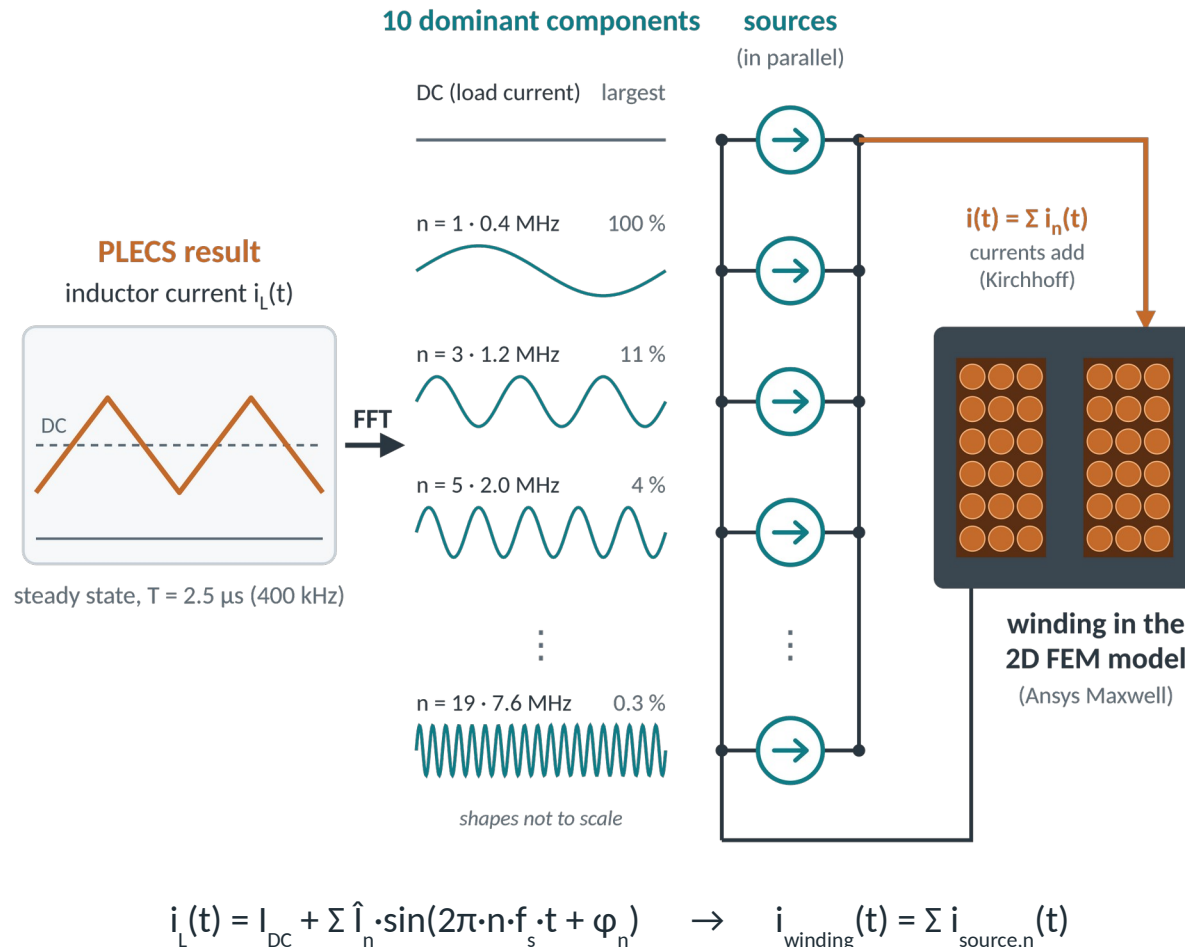
A one-way chain: the circuit simulation supplies the current, the FEM turns it into losses



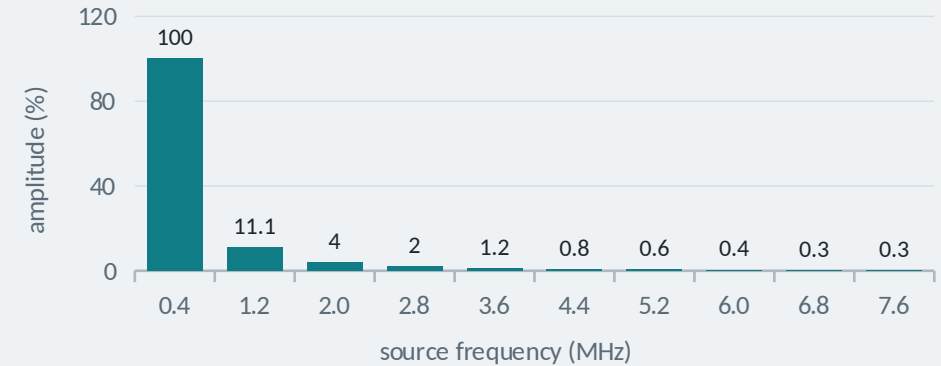
No recursion in the poster's workflow. Information flows forward only: PLECS → FFT → FEM → losses. Nothing from the FEM is fed back into PLECS; the chain runs once per design (solid wire, litz wire).

From PLECS waveform to FEM excitation

Fourier series: the current is split into sinusoids, and parallel sources add them back up in the FEM



Amplitudes of the 10 ripple components



Why sinusoids?

Skin and proximity effect depend on frequency. Split into sinusoids, the FEM sees exactly the frequency content the winding carries.



Why in parallel?

Parallel current sources add their currents (Kirchhoff), so the winding carries $i(t) = \sum i_n(t)$: the PLECS current.



Why only 10?

Amplitudes fall as $1/n^2$: 10 components rebuild the ripple within ~2 %. More would add higher frequencies (7.6 MHz: $\delta \approx 24 \mu m$ in copper), i.e. finer mesh and smaller time steps.

Inside the FEM — and when a loop would be needed

The FEM turns fields into losses once per design; iterating only pays off when the results change the inputs

From fields to losses

● Winding (copper)

$$P_w = (I/T) \cdot \iint J^2 / \sigma \, dA \, dt = \sum_{ac} R_{ac} (f_n) \cdot I_{n,rms}^2$$

J comes from the FEM and already contains skin, proximity and air-gap fringing effects. This is where solid and litz wire differ.

● Core (ferrite)

$$P_{core} = \int p_v \, dV, \text{ e.g. } p_v = k \cdot f^\alpha \cdot \hat{B}^\beta$$

The loss density follows from the local flux density B in every mesh element and the material's loss data (Steinmetz-type).

● 2D model

3D loss $\approx$ 2D loss per metre $\times$ effective length

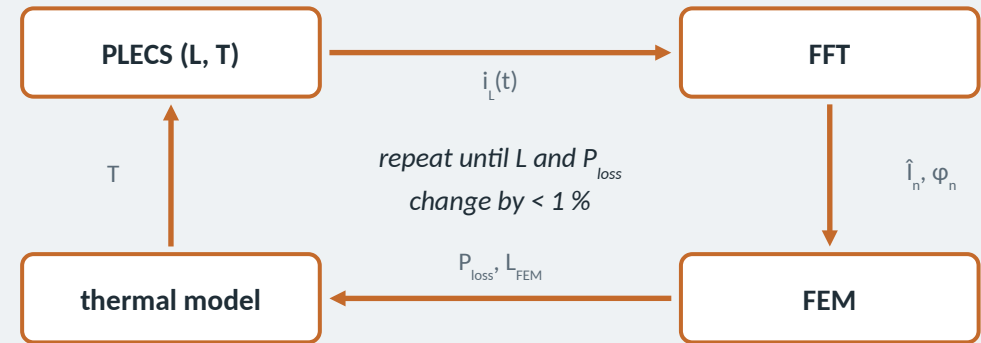
A 2D cross-section is fast but does not resolve end turns and corners of the RM core.

One-way chain vs. iterative (recursive) loop

Poster: one-way, run once per design



General method, not used on the poster: iterate until nothing changes

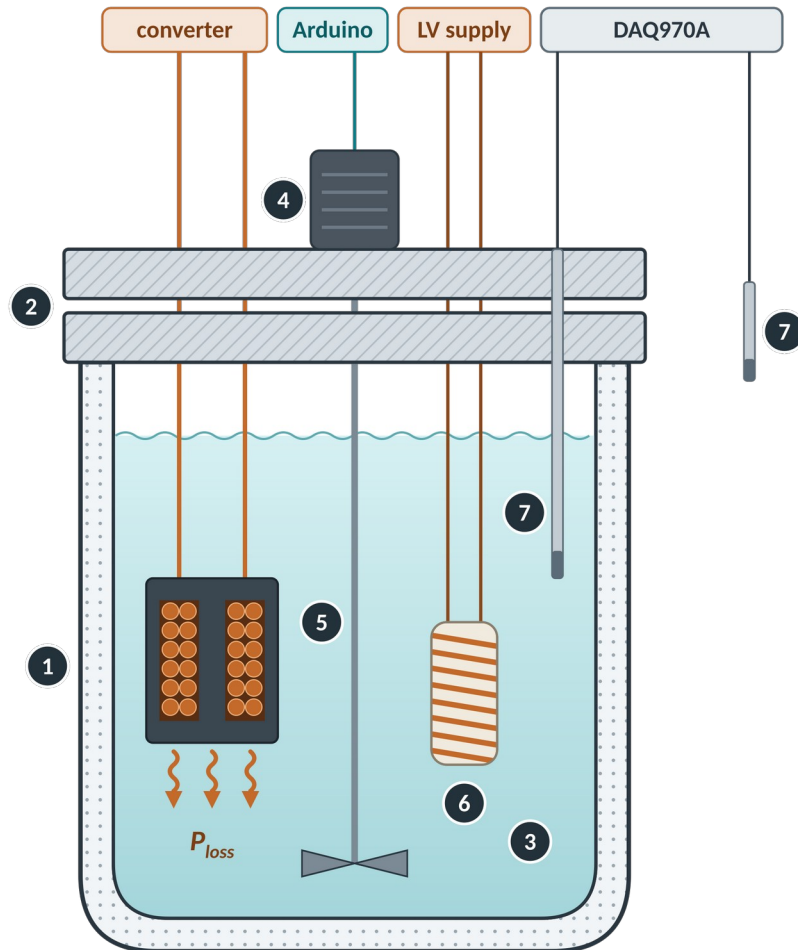


A loop is only needed when the results change the inputs:

- the inductance from the FEM (air gap, DC bias, saturation) differs from the L used in PLECS $\rightarrow$ different ripple
- self-heating raises the copper resistance (+0.39 %/K) and changes the ferrite loss $\rightarrow$ different losses

Thermal side: measuring the losses as heat

Near-adiabatic calorimeter: the inductor's losses heat a stirred liquid of known heat capacity



$$P_{\text{loss}} = C_{\text{Th}} \cdot dT/dt \approx C_{\text{Th}} \cdot \Delta T / \Delta t$$

C_{Th} : total heat capacity inside the adiabatic enclosure; ΔT : its temperature change during the time Δt in which the power is dissipated. Valid because almost no heat leaves the Dewar.

Set-up

- 1 Dewar jar (vacuum-insulated)
- 2 Custom double lid
- 3 Insulating liquid, stirred
- 4 Stepper motor + Arduino (stirrer)
- 5 DUT: inductor, fully submerged
- 6 LV supply + wire-wound resistor
- 7 PT100 (fluid, ambient) → DAQ970A

Why this design?



Losses become heat

Core and winding losses are captured together, independent of waveform and frequency [3, 6].



Near-adiabatic

Unlike other calorimetric methods, no mass-flow measurement is necessary [4, 5].



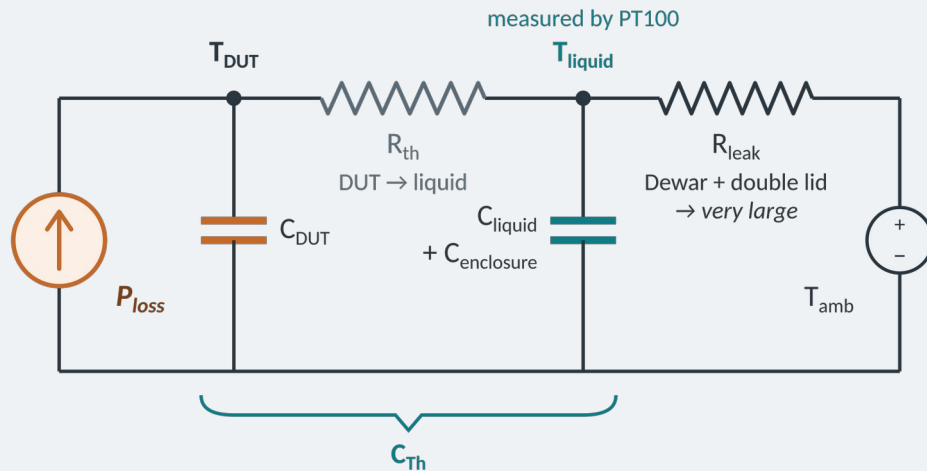
Stirred liquid

Good electrical insulation and high heat capacity at reasonable cost; stirring heats it uniformly.

Thermal model and calibration

Known heater powers turn the temperature slope into watts — under the same thermal conditions as the DUT

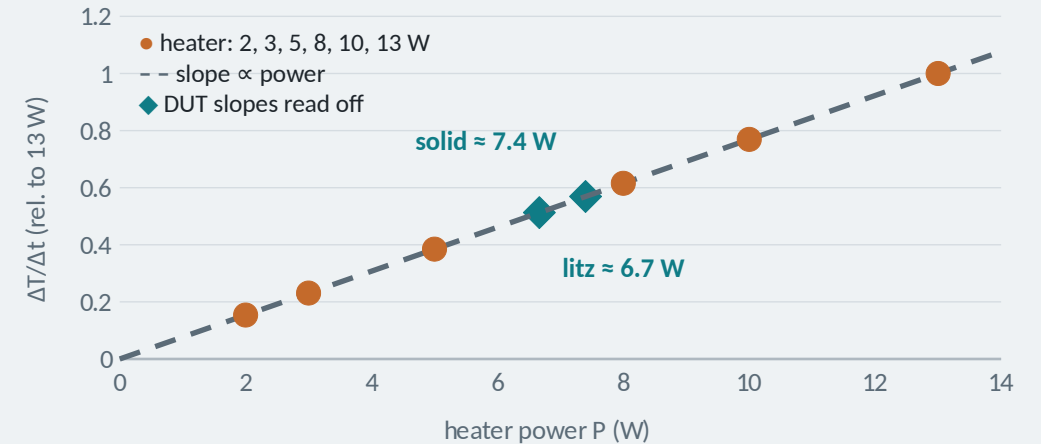
Lumped thermal model of the calorimeter



$$C_{Th} \cdot dT/dt = P_{loss} - (T - T_{amb})/R_{leak} \rightarrow P_{loss} \approx C_{Th} \cdot \Delta T/\Delta t \text{ for large } R_{leak}$$

- C_{Th} includes the DUT → it is fully submerged also during calibration
- R_{th} delays the response → only the most linear part of $\Delta T(t)$ is used
- Leakage is small, not zero → every run starts at 35 °C
- One liquid temperature → continuous stirring keeps the liquid uniform

Calibration: slope $\propto$ power

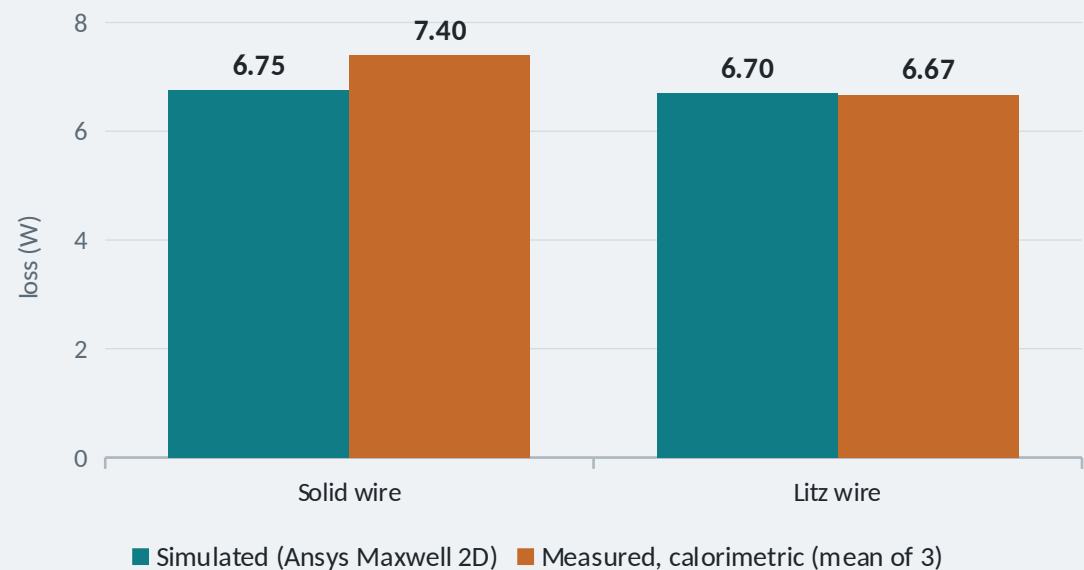


- 1 Calibrate:** LV supply + wire-wound resistor at known power (4-wire measurement)
- 2 Fit:** most linear region of each $\Delta T(t)$ record → one $\Delta T/\Delta t$ slope per power
- 3 Measure:** DUT operated as in normal operation → its $\Delta T/\Delta t$ slope
- 4 Interpolate:** loss read between the calibrated slopes

Results: simulated vs. thermally measured losses

Litz-wire inductor agrees within 1.5 %; the solid-wire inductor is measured 8–11 % above simulation

Inductor loss — RM14LP, 400 kHz, ≈ 880 W



Wire	Run 1	Run 2	Run 3	Mean	Simulated	Meas. vs. sim.
Solid	7.3	7.4	7.5	7.40	6.75	+9.6 %
Litz	6.7	6.7	6.6	6.67	6.70	-0.5 %

Six calorimetric runs from the poster, values in W. Deviation = (measured mean – simulated) / simulated.

LITZ WIRE

-0.5 %

measured vs. simulated (mean)
per run: -1.5 % ... 0.0 %

SOLID WIRE

+9.6 %

measured vs. simulated (mean)
per run: +8.1 % ... +11.1 %



Accurate loss prediction

The workflow predicts the losses of the litz-wound inductor very close to the thermally measured values.



Solid-wire effect under-represented

Measured: solid is 0.73 W above litz; simulated: only 0.05 W. Why this occurs is left for future analysis [10].



Good approximation for inductor design

Measured losses are within ~11 % of the simulated losses; the calorimeter resolves 0.1 W differences between runs.

References

The poster, the reference it cites, and related work on thermal loss measurement and magnetic loss modelling

The poster and the reference it cites

- [1] P. Skoff, M. Stoiber, J. Reynvaan, and W. Konrad, "Experimental verification of automated loss-simulation of magnetic components," poster, *Power Electronics for Energy Transition Symposium*, Vienna, Austria, Sep. 29–30, 2026.
- [2] J. Reynvaan, M. Pajnić, and J. Krenn, "Evaluating fluid based transient calorimetric method for measurement of the ferrite core losses," in *Proc. ICPE 2023-ECCE Asia*, Jeju, Korea, 2023, pp. 3308–3313, doi: 10.23919/ICPE2023-ECCEAsia54778.2023.10213808.

Thermal (calorimetric) loss measurement – similar concept

- [3] P. Papamanolis, T. Guillod, F. Krismer, and J. W. Kolar, "Transient calorimetric measurement of ferrite core losses up to 50 MHz," *IEEE Trans. Power Electron.*, vol. 36, no. 3, pp. 2548–2563, 2021.
- [4] D. Christen, U. Badstuebner, J. Biela, and J. W. Kolar, "Calorimetric power loss measurement for highly efficient converters," in *Proc. IPEC 2010 (ECCE Asia)*, Sapporo, Japan, 2010, pp. 1438–1445.
- [5] S. Weier, M. A. Shafi, and R. McMahon, "Precision calorimetry for the accurate measurement of losses in power electronic devices," *IEEE Trans. Ind. Appl.*, vol. 46, no. 1, pp. 278–284, 2010.
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Magnetic loss and thermal modelling / simulation – similar concept

- [9] P. L. Dowell, "Effects of eddy currents in transformer windings," *Proc. IEE*, vol. 113, no. 8, pp. 1387–1394, 1966.
- [10] R. P. Wojda and M. K. Kazimierczuk, "Winding resistance and power loss of inductors with litz and solid-round wires," *IEEE Trans. Ind. Appl.*, vol. 54, no. 4, pp. 3548–3557, 2018.
- [11] J. Mühlethaler, J. Biela, J. W. Kolar, and A. Ecklebe, "Improved core-loss calculation for magnetic components employed in power electronic systems," *IEEE Trans. Power Electron.*, vol. 27, no. 2, pp. 964–973, 2012.
- [12] P. A. Kyaw, M. Delhommais, J. Qiu, C. R. Sullivan, J.-L. Schanen, and C. Rigaud, "Thermal modeling of inductor and transformer windings including litz wire," *IEEE Trans. Power Electron.*, vol. 35, no. 1, pp. 867–881, 2020.
- [13] N. Förster et al., "An open-source FEM magnetic toolbox for calculating electric and thermal behavior of power electronic magnetic components," in *Proc. EPE'22 ECCE Europe*, Hanover, Germany, 2022.
- [14] N. Djekanovic, M. Luo, and D. Dujic, "Integrated simulation approach to loss calculations of power converter systems," in *Proc. PCIM Europe digital days*, 2020, pp. 418–425.
- [15] T. Guillod, P. Papamanolis, and J. W. Kolar, "Artificial neural network (ANN) based fast and accurate inductor modeling and design," *IEEE Open J. Power Electron.*, vol. 1, pp. 284–299, 2020.

HOW THE REFERENCES CONNECT TO THE SLIDES

Thermal measurement (slides 6–7): [2] same group's fluid calorimeter, [3] transient calorimetry, [4]–[8] other loss-measurement methods

Magnetic simulation (slides 2–5): [9]–[11] winding and core loss models, [13]–[15] automated FEM workflows

Litz vs. solid wire (slide 8): [10] winding loss, [12] winding thermal model